

AI3403 – Multi-Agent Systems

Multi-Agent Wildfire Search, Survivor Tracking and Adaptive Rescue Coordination

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1. Motivation and Problem Statement

Wildfire search and rescue is a dynamic MAS problem where survivor locations, fire conditions, communication links, and rescue resources change continuously. We propose a decentralized MAS in which heterogeneous UAVs cooperatively search for survivors, track their movement, and coordinate limited rescue resources while satisfying safety requirements. The scenario begins when a person travelling on a nearby road reports a wildfire. Emergency authorities reach the forest access point and use parked vehicles, trail/entry records, and approximate group information to infer that visitors may still be inside. Their current locations are unknown, so UAVs are deployed to search the likely regions.

The main objective is to study how decentralized coordination can improve survivor localization, tracking, and rescue when the fire spreads dynamically, communication topology changes, agents fail, and rescue capacity is limited.

2. MAS Formulation

The system contains four interacting components:

- **UAV agents:** Heterogeneous UAVs differ in speed, battery, sensing capability, and communication range. They communicate through a dynamic UAV network whose links can appear or disappear as vehicles move.
- **Survivor agents:** Survivors are independent behavioural agents with different health, mobility, route knowledge, forest familiarity, situational awareness, and risk preference. Nearby survivors communicate by voice, allowing groups to form or fragment as they move.
- **Wildfire environment:** The forest contains terrain, vegetation and trails. Fire propagation changes with wind, terrain and vegetation, producing a continuously changing risk map that affects both survivors and UAVs.
- **Rescue resources:** Ground teams and, when available, helicopters have limited availability, capacity and response time. Resources are assigned according to survivor priority, estimated response time, risk and capacity.

The survivor and UAV communication networks are separate. Survivors do not directly communicate with UAVs; UAVs locate them through sensing.

3. Proposed Approach

Search and sensing: UAVs search the estimated regions using a probabilistic sensing model. Detection probability depends on distance, smoke, vegetation, terrain and UAV sensing capability.

Tracking and handoff: After detection, a UAV maintains an estimate of the survivor's location and tracking confidence. If tracking is lost or another UAV is better positioned, the task is handed over using the information available through the UAV network.

Adaptive task allocation: RL will be used for high-level UAV task allocation. Possible tasks include SEARCH, TRACK, HANDOFF, SUPPORT RESCUE and RETURN. The RL agent decides what each UAV should do, while risk-aware path planning determines how it reaches the selected task.

Distributed coordination: UAVs exchange task and survivor information with connected neighbours. Communication disconnections and UAV failures can therefore produce incomplete or delayed information, requiring the remaining UAVs to reallocate tasks.

Rescue allocation: Rescue resources will be assigned using an optimization-based approach considering ETA, risk, survivor priority and resource capacity. After a resource reaches a survivor, the rescue is confirmed and the information is propagated through the connected UAV network.

4. Safety and Robustness

The project will study MAS operation under changing communication topology, UAV failures, survivor-group fragmentation and changing fire conditions. Disconnected UAVs will continue using their local information and synchronize when connectivity is restored.

STL will be used to express safety and mission requirements such as:

- maintaining safe separation between UAVs;
- avoiding high-risk fire regions;
- revisiting important survivors within specified time limits;
- completing high-priority rescue actions within deadlines; and
- stopping redundant search after rescue confirmation.

5. Two-Month Implementation Plan

The project will be developed in stages, with each stage producing a working component for integration into the final system.

1. Weeks 1–2: Environment

Build the forest map, terrain, vegetation, trails, wind model, wildfire propagation and fire-risk estimation.

2. Weeks 2–3: UAVs

Implement heterogeneous UAVs, battery states, movement, sensing, communication range and dynamic UAV connectivity.

3. **Weeks 3–4: Survivors**

Implement heterogeneous survivor behaviour, movement, health, mobility, route knowledge, voice-based communication and group fragmentation.

4. **Weeks 4–5: Search and tracking**

Integrate survivor detection, tracking, information sharing and UAV task handoff.

5. **Weeks 5–6: RL coordination**

Develop and train the RL-based UAV task allocation policy and compare it with simple non-learning strategies.

6. **Weeks 6–7: Rescue and failures**

Implement rescue-resource allocation, rescue confirmation, information propagation, communication failures and UAV failures.

7. **Weeks 7–8: Integration and evaluation**

Complete the simulation and dashboard, conduct experiments, generate plots, analyse results and prepare the final demonstration.

6. **Demonstration and Evaluation**

The final demonstration will be a real-time 2-D simulation of a forest. The main window will display the fire front, fire-risk regions, trails, UAVs, survivors, search regions, communication links and rescue teams.

A dashboard will show UAV battery and current task, survivor status and tracking confidence, rescue assignments, and UAV-network connectivity.

The demonstration will show the complete mission:

Wildfire Report → Search → Detection → Tracking → Task Reallocation → Rescue Assignment → Rescue Confirmation

The system will be tested under changing fire conditions, survivor movement, survivor-group fragmentation, communication disconnections, UAV failures, sensing difficulty and limited rescue resources.

The proposed method will be compared with random, nearest-agent and greedy allocation strategies. Evaluation metrics will include detection time, tracking loss, rescue time, rescue success rate, survivor survival rate, battery consumption, communication overhead, task reallocations, safety violations and scalability.

7. **Individual Contributions**

Member	Primary contribution
Akshay	Environment: terrain, vegetation, trails, wind-dependent fire propagation, fire-risk model and scenario generation.
Krishna chaitanya	UAVs: heterogeneous UAV models, battery, movement, sensing, path planning, survivor tracking and task handoff.
Preetham	Survivors: survivor behaviour, movement, health and mobility, route knowledge, voice communication and group fragmentation.
Sukesh	Coordination and RL: UAV communication topology, information sharing, task allocation, RL training and failure handling.
Vishal	Rescue and evaluation: rescue allocation, rescue confirmation, STL safety, experiments, metrics, dashboard and result analysis.

All members will contribute to integration, experiments, analysis, documentation and the final presentation in addition to their primary modules.

Expected Outcome: A working decentralized MAS that demonstrates cooperative UAV search, survivor tracking, adaptive task allocation, rescue coordination, failure handling and safety-aware operation in a dynamic wildfire environment.